

VATSAL SANDEEP NAIK

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EXPERIENCE

Northeastern University

Boston, MA

Graduate Teaching Assistant

Jan 2026 – Apr 2026

- Mentored 50+ engineers on software quality, **API design patterns**, testing practices, and deployment workflows translating complex engineering concepts for both technical and non-technical audiences.
- Built **LLM-powered developer assistant** (Anthropic API, **MCP**) automating Q&A workflows for 50+ engineers in production, with context-aware tool invocation and hallucination guardrails.

DMSB AI Strategic Hub

Boston, MA

Machine Learning Engineer

Jan 2025 – Dec 2025

- Developed a 4-agent RAG system (LangChain, FAISS, Hugging Face embeddings) for automated academic submission grading, across 200+ submissions per cycle with custom RAGAS metrics and hallucination detection.
- Built and maintained **CI/CD workflows** for model experimentation, A/B testing **4 fine-tuning configs** (LoRA, QLoRA, PEFT) with automated evaluation gates to select and promote the production model configuration.

Gupshup.ai

India

Software Engineer

Dec 2021 – Jul 2024

- Built **Node.js** and **Express.js** microservices on **AWS EKS** processing 1M+ messages/day, handling **intent routing**, chatbot agent orchestration, and **rich message delivery** across **Meta platforms** for clients in BFSI and fintech.
- Designed and extended Gupshup's **shared RESTful API framework** by adding integrations for **NLP, OCR, Razorpay, Stripe**, and **META Catalog**, powering complete e-commerce bot workflows that were deployed in **25+ client projects**.
- Architected a **bot workflow solution** (MongoDB, Redis, Dialogflow) that unified core conversational logic as a base layer, reducing the effort for new integrations (payment, logistics APIs) from **2 days to 4 hours**.
- Developed **Kafka** async pipeline with dead-letter queues decoupling message ingestion from downstream processing, reducing delivery failures on high-traffic client deployments.
- Automated build and deployment workflows using **Jenkins** and **Bash** scripts, eliminating manual coordination across **3 environments** and reducing release cycles for client bot deployments.

Software Engineer Intern

Apr 2021 – Nov 2021

- Integrated Gupshup's messaging APIs with **Salesforce** and **Zoho** CRMs, cutting integration turnaround by **~30%** via Python and Postgres based workflows.
- Debugged live **OAuth misconfigurations** and **webhook schema failures** for enterprise accounts, restoring **same-day API uptime** and shipping automated health checks to prevent recurrence.

PROJECTS

SimpliCloud – Cloud-Native E-Commerce Backend | AWS, Terraform, Kinesis, S3, Kubernetes, CloudWatch

[GitHub](#)

- Provisioned production-grade AWS platform via **Terraform IaC** with multi-AZ **VPC**, **EKS** cluster, **ElastiCache Redis**, **OpenSearch**, **WAF**, **CloudFront**, **Kinesis** streams, **X-Ray** tracing, and **CloudWatch** observability.

Cost Sherlock – Multi-Agent AI Code Analysis Pipeline | Python, LangChain, RAG, Claude API

[Live Demo](#)

- Architected a **4-agent AI pipeline** with multi-step reasoning, task decomposition, and **RAG**-based retrieval using LangChain; built **RAGAS eval layer** with hallucination detection and output guardrails.

JobLens – Full-Stack AI Developer Platform | Python, FastAPI, PostgreSQL, Redis, React, AWS

[Live Demo](#)

- Shipped end-to-end **developer-facing platform** with **FastAPI** backend, async workers, **PostgreSQL** and **Redis** data layer, and **React** frontend on **AWS** with **Prometheus/Grafana** observability.

TECHNICAL SKILLS

Languages: Python (primary), TypeScript, JavaScript, C++, Go, Java, SQL, Bash

Backend: Node.js, Express.js, FastAPI, REST, Microservices, PostgreSQL, Kafka, Redis, MySQL, MongoDB, RabbitMQ

AI/ML: Pytorch, TensorFlow, Pandas, Numpy, MCP, Claude Code, Prompt Engineering, LangGraph, RAG, MLflow

Cloud: AWS (EC2, S3, RDS, Lambda, VPC, CloudWatch, API Gateway, DynamoDB, Kinesis), Kubernetes, Docker, Serverless

DevOps: GitHub Actions, Jenkins, Git, Packer, Ansible, Terraform, Prometheus, Grafana, Katalon, Linux, Postman

EDUCATION

Master of Science, Information Systems | Northeastern University

May 2026

Bachelor of Science, Information Technology | University of Mumbai

May 2021